

ANDREW GEORGIU

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Programming Experience: Frameworks & Tools: Cloud Platforms & Data: Languages:

Go, C++, Python, TypeScript, Ruby, Java, SQL, JavaScript
PyTorch, TensorFlow, scikit-learn, ReactJS, gRPC/Twirp, Kafka, Docker, Kubernetes
Azure, AWS, Google Cloud (GCP), CosmosDB, MySQL, Kusto, Apache Beam, Hadoop
English (Native), Greek (Conversational)

EXPERIENCE

Software Engineer - GitHub (Copilot)

Dec. 2024 - Present

- **Agent Automations (DRI).** Built GitHub's scheduled and event-driven coding agent: trigger system, distributed scheduler and per-automation billing. 870 automations created in the first three days.
- **Copilot Mission Control.** Core engineer on the agentic platform behind Copilot, at 64k monthly active developers and 100k agent sessions per day. Shipped agent icebreakers and reasoning-effort control across GitHub.com agent surfaces.
- **Real-time session events.** Replaced polling with event-driven Alive subscriptions for agent sessions, cutting client requests by ~96%, and designed a refresh-token architecture that reduced OAuth record creation ~24x and removed mid-session authentication failures.
- **Security champion.** Cleared 100% of the platform's overdue security SLAs, remediated an IDOR in the agent session API, and scoped agent write access to mitigate prompt-injection risk.
- **Prompting & observability.** Created the Copilot prompt builder system to cut per-request token consumption, and created LLM Tracing, giving engineers shareable end-to-end visibility into Copilot prompts.
- **Community.** Founded the LLM Research Club and wrote the AI Excellence Program, a 9-module LLM curriculum, alongside 50+ weekly research digests.

Software Engineer - Google

Sep. 2022 - Aug. 2024

- **Data Flow Graph.** Engineered and optimised C++ MapReduce feeds processing trillions of Google traffic requests daily, driving large-scale data processing for privacy and AI within Protected Data.
- **Ingestion latency.** Launched a feed creation system that cut data latency from 72 hours to under 24, streamlining over 120 billion graph edges per day. Received a performance award for the efficiency gains.
- **DMA compliance.** Introduced a Queries-Per-Second metric to evaluate graph coverage, now relied on by 50+ DMA-ready teams at Google for debuggability under the Digital Markets Act.
- **ML for debugging.** Formed a global team to build a Message Passing Neural Network in TensorFlow that identified access violations, cutting average debugging time from 1 hour to 5 minutes.

Software Engineer Intern - Barclays

Summers 2020 & 2021

- **Payments ML.** Built a machine learning model for the Global Payments - Faster Payments team that identified infrastructural requirements from processing payment volumes, improving the team's ability to handle rising transaction volumes.
- **Forecasting & visualisation.** Designed a seasonal regression model in scikit-learn to forecast payment trends alongside scalable capacity insights, which the team used to automate requests for additional cloud capacity, and built a React platform graphing volumes against the model's predictions.
- **Cloud security.** Implemented mutual authentication across AWS EC2 using Ruby Chef cookbooks with the IBM CMOD cloud team, and delivered a cloud architecture training series covering enterprise-scale AWS operations.

Freelance Full Stack Engineer - LouGeo Ltd.

Jan. 2019 - Jun. 2020

- Built a ReactJS and Firebase stock management system for Yiamas Greek Taverna, plus a NextJS site with an integrated booking system that increased reservations by over 50% within six months of launch.
- Built a ReactJS site with an Express.js backend for music artist Giancarlo Canetti, reaching over 50,000 listeners independently of major music platforms.

EDUCATION

University of Cambridge

Sep. 2021 - Jun. 2022

Advanced Computer Science, MPhil - 73% Average.

- Awarded **The TPP Alan Turing Scholarship** from King's College, Cambridge.
- Hyper-network research into updating implicit knowledge inside pre-trained large language models without expensive and unstable fine-tuning, applied to fact verification and explainable AI. Papers at A-Georgiou/Research.Papers.

University of Liverpool

Sep. 2018 - Jun. 2021

Computer Science, BSc - First Class Honours: 84% Average

- **Andrew Young Prize, 2021** Awarded for best overall performance in Final Year Honours examinations.
- **BCS Merseyside Prize, 2021** Awarded for being the best overall student on a degree programme.
- **Microsoft Student Ambassador** and **Google DSC Lead**, running workshops and community-driven projects.

PROJECTS

SmartDNN - A High-Performance C++ Deep Learning Library

2024

github.com/A-Georgiou/SmartDNN

- Dependency-free neural network library using template meta-programming for type-safe tensor operations with automatic precision promotion.
- SIMD-parallelised compute kernels, reducing MNIST training from 83 minutes to 10 seconds per epoch.